

BPMN Modeling Fundamentals

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[BPMN diagram tutorial](#)



Business Process Model and Notation (BPMN) uses globally standardized symbols to illustrate complex business processes with clarity. Unlike basic flowcharts, BPMN explicitly details triggers, roles and responsibilities, decision branching/merging, and whether tasks occur sequentially or simultaneously.

1. Core Building Blocks & Modeling Rules

○ Events

- Definition: Events represent things that happen during a process. They mark when a process starts, shifts, or ends.
- Visual Representation: Shown as circles:
 - Start Event: Circle with a single thin line (e.g., *"Customer wants a drink"*).
 - Intermediate Event: Circle with a double line (marks a shift, trigger, or waiting period).
 - End Event: Circle with a thick solid line (e.g., *"Customer receives drink"*).
- Key Rules:
 1. Every BPMN diagram must include at least one start event and one end event.
 2. Events are not actions performed; they simply happen or act as markers.

→ Sequence Flows

- Definition: Sequence flows establish the order in which tasks, events, and decisions occur, guiding the path from start to finish.
- Visual Representation: Shown as an arrow connecting elements.
- Key Rules:
 1. Every BPMN diagram must have at least one sequence flow.
 2. Sequence flows always indicate the exact direction of the process.

□ Activities (Tasks)

- Definition: Activities represent the actual work or actions required to move a process forward. They require effort from a person, a system, or another process.
- Visual Representation: Shown as rounded rectangles.
- Examples: *"Insert Money"*, *"Select Drink"*, *"Machine Dispenses Drink"*, *"Check Money"*.
- Key Rules:
 1. Activities always represent active work that pushes the process toward completion.
 2. Activities must be connected within the sequence flow.

◇ Gateways

- Definition: Gateways direct process flow at points where paths split, merge, or take different directions based on specific conditions. They control the flow but do not perform work themselves.
- Visual Representation: Shown as diamond shapes.
- Three Primary Types:
 1. Exclusive Gateway: Only one path can be taken based on conditions (e.g., *"Is the drink in stock?"* → Yes / No).
 2. Inclusive Gateway: One or more paths may be taken depending on conditions.

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3. Parallel Gateway: All paths are executed simultaneously (typically used in complex processes).

- Key Rules:

1. Gateways always indicate a split or a merge in flows.
2. Gateway symbols specify how many paths may be taken.

❑ Pools and Swim Lanes

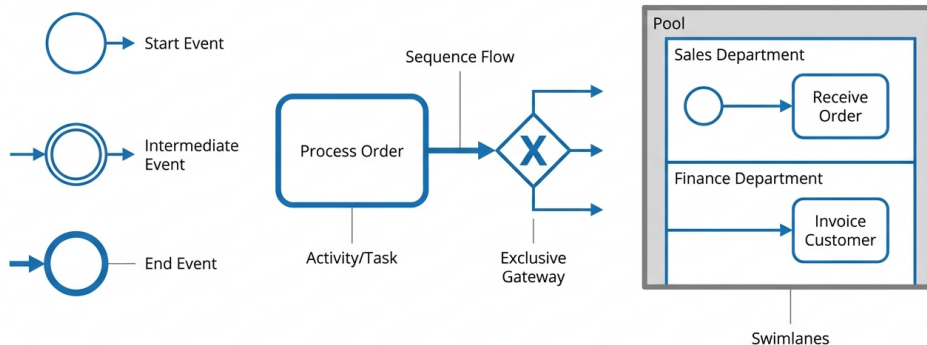
- Definition: Essential structures that establish organizational clarity, define accountability, and highlight gaps in responsibility.
- Pool: A large rectangle representing the entire overall process being modeled (e.g., "Ordering a drink from a vending machine").
- Swim Lanes: Subdivisions inside a pool showing who or what (person, system, role) is responsible for each specific action.
- Key Rules:
 1. A pool represents the overall process.
 2. Swim lanes define responsibility, and each lane should correspond to only one role.
 3. Sequence flows should cross lanes neatly.

2. Adding Precision & Advanced Context

Once foundational elements are established, additional detail can be incorporated for greater clarity:

- Task Types: Add icons inside activity boxes (e.g., a hand icon for manual tasks) to denote physical actions.
- Timers: Include intermediate timer events to indicate waiting periods or delays within the process flow.

CORE BUSINESS PROCESS MODEL AND NOTATION (BPMN 2.0) SYMBOLS



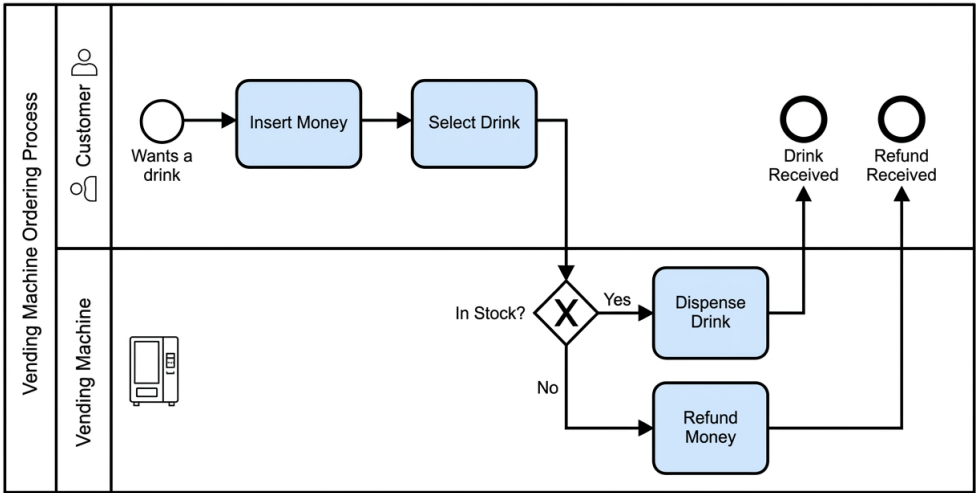
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3. Quick Reference Checklist of BPMN Rules

Element	Icon Shape	Core Function	Mandatory Rule
Start / End Events	Circle (Thin / Thick)	Mark start/end triggers	Must have at least one Start and one End event
Sequence Flow	Arrow	Show order and direction	Must have at least one sequence flow showing direction
Activity	Rounded Rectangle	Represent work performed	Must represent work and connect to sequence flows
Gateway	Diamond	Direct branching/merging	Always indicates splits or merges of paths
Pool & Swim Lanes	Outer Box & Subdivisions	Map roles & accountability	Each lane must represent exactly one role

Example:

Vending Machine



Gemini Notebook

BPMN Web Tutorial
<https://lucid.co/diagram/bpmn/tutorial>